SKILL BUILDERS · UNIT 1 · SCORE 3

Real Numbers: Practice Set

NC.8.NS.1–2 · NC.8.EE.2

ANSWER KEY

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.NS.1

Is $0.1212\ldots$ rational or irrational?

- **(A) Rational**
- (B) Irrational

2 Score 3 NC.8.NS.1

Is $\sqrt{144}$ rational or irrational?

- **(A) Rational**
- (B) Irrational

3 Score 3 NC.8.NS.1

Is $\sqrt{8}$ rational or irrational?

- **(A) Irrational**
- (B) Rational

4 Score 3 NC.8.NS.1

Write $0.\overline{7}$ as a fraction.

- **(A) 7/9**
- (B) 7/10
- (C) 1/7
- (D) 7/99

5 Score 3 NC.8.NS.1

Write $0.\overline{45}$ as a fraction in simplest form.

- (A) 45/100
- **(B) 5/11**
- (C) 45/99
- (D) 9/20

1. $0.\overline{45} \rightarrow 45/99$
2. $45 \div 9 = 5,\ 99 \div 9 = 11$

↓ simplify

3. $= 5/11$

6 Score 3 NC.8.EE.2

Find $\sqrt{121}$.

Answer: **11**

1. $11 \times 11 = 121$
2. $\sqrt{121} = 11$

7 Score 3 NC.8.EE.2

Find $\sqrt[3]{64}$.

Answer: **4**

1. $4 \times 4 \times 4 = 64$
2. $\sqrt[3]{64} = 4$

8 Score 3 NC.8.NS.2

$\sqrt{40}$ is between which two whole numbers?

- (A) 20 and 21
- **(B) 6 and 7**
- (C) 5 and 6
- (D) 36 and 49

9 Score 3 NC.8.NS.2

Estimate $\sqrt{50}$ to the nearest tenth.

Answer: **7.1**

1. $49 < 50 < 64 \rightarrow 7 < \sqrt{50} < 8$
2. $(50 - 49) \div (64 - 49) = 1 \div 15 \approx 0.07$
3. $\sqrt{50} \approx 7.1$

10 Score 3 NC.8.NS.2

Order from least to greatest:

$\pi, \sqrt{10}, 10/3, 3.5$

(A) √10, π, 10/3, 3.5

(B) π, √10, 10/3, 3.5

(C) 3.5, 10/3, √10, π

(D) π, 10/3, √10, 3.5

1. $\pi \approx 3.14, \sqrt{10} \approx 3.16, 10/3 \approx 3.33, 3.5 = 3.50$
2. $\pi < \sqrt{10} < 10/3 < 3.5$

11 Score 3 NC.8.EE.2

Find the mistake. Sam wrote: $\sqrt{36} = 18$. What went wrong?

(A) Sam divided 36 by 2. √36 = 6 because 6 × 6 = 36.

(B) Sam is right: 18 + 18 = 36.

(C) √36 = 9 because 4 × 9 = 36.

(D) √36 = 72 because 36 × 2 = 72.